ANUBUTHI KOTTAPALLI

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EDUCATION

University of Chicago — **Sep 2024 - Dec 2025**
Master's, Applied DataScience — *GPA: 4*
- Coursework: Big Data, Bayesian ML, Time Series Analysis, NLP, Algorithmic Marketing, Generative AI, Causal Models

PES University
Bachelor's, Computer Science — *GPA: 3.92*
- Coursework: Data Analytics, Machine Intelligence, Network Analysis, DBMS, Data Structures, Graph Theory, Cloud Computing

SKILLS

Languages & Libraries: Python, (NumPy, Pandas, TensorFlow, Keras, PySpark, NLTK, Torch, statsmodels), SQL, R
ML & AI: Predictive Modelling, NLP, A/B Testing, Gen AI, RAG, Forecasting, Recommender Systems (Collaborative Filtering, KNN)
Data Tools & Platforms: Excel, Jupyter, GCP, Docker, Hadoop, Streamlit, Gradio, Git, Apache Spark
Data Visualization: Power BI, Tableau, Excel, Matplotlib, Seaborn, Plotly
Algorithmic Marketing: Conjoint Analysis, Competitive Analytics, Promotional & Pricing Analysis, Market Segmentation, Market Basket
Core Competencies: Strategic thinking, Empathy, Teamwork, Creativity, Quick Learner, Communications, Problem Solving, Active Listener

EXPERIENCE

Argonne National Laboratory — **Chicago, IL, USA**
Part-Time Capstone Research—Connected & Autonomous Vehicle(CAV)Systems — *Apr 2025 - Present*
- Developed and optimized a computer vision product to estimate lead vehicle distance in sensor-limited autonomous systems, using Vision Transformers , Neural Networks and CNN based architectures.
- Applied YOLO for object detection and MiDaS for monocular depth estimation, leveraging advanced mathematics and image preprocessing techniques to obtain features used for distance estimation.
- Designed and rigorously evaluated over 1,000 simulated data in CARLA, applying computer vision models (Vision Transformers, CNNs) to estimate lead vehicle distance and improve autonomous user safety.

Evoke Technologies — **Dayton, OH, USA**
Data Science Intern — Capital Markets, AI & Financial Research — *Jun 2025 - Sep 2025*
- Drove data-driven decision making by delivering weekly equity research reports to senior analysts, providing strategic recommendations that directly influenced portfolio positioning.
- Applied advanced analytical methods and XGBoost/LightGBM models to classify trading anomalies and generate predictive signals, demonstrating strong skills in modeling and feature engineering from complex, time-series data.
- Automated an ingestion pipeline to process 9,000+ daily stock trades and analyst estimates using GCP Cloud Functions, reducing manual processing time by over 5 hours a day.

GlaxoSmithKline — **Bengaluru, KA, India**
Graduate Intern —Cyber Analytics & Detection Engineering — *Jan 2024 - Jul 2024*
- Reduced analyst workload by over 8 hours per day by developing a proof of concept to automate the service ticket triage process.
- Created 20+ threat detection playbooks by collaborating with cross-functional teams to design ML-driven detection workflows, mapping Office 365 telemetry to the MITRE ATT&CK framework.
- Transformed raw user and entity behavior data (UEBA) into an intuitive visual story using Power BI, enabling analysts to spot anomalous patterns and make faster, insight-driven decisions.

PROJECTS

Uncertainty-Aware Classification - *Link to project* — **Chicago, IL, USA**
Student — *Jul 2025 - Aug 2025*
- Developed and optimized a Bayesian Deep Learning framework for multi-class classification of Diabetic Retinopathy (DR) using retinal fundus images, experimenting with four model variants including ResNet18 and Custom CNN architectures.
- Implemented Monte Carlo Dropout and Variational Inference to robustly quantify and estimate predictive uncertainty (via entropy), enhancing model interpretability in a critical medical context.
- Engineered an uncertainty-aware decision-making mechanism using class-specific entropy thresholds to allow the model to abstain from high-uncertainty predictions, achieving the crucial clinical goal of reducing false negatives for high-risk DR stages (3–4).

Will Turing Bots Replace Human Developers? —Big Data Analytics - *Link to project* — **Chicago, IL, USA**
Student
- Processed and analyzed 1.3 TiB of GitHub Archive data using PySpark on GCP to extract behavioral time series and code-level metrics across 100K+ repositories.
- Uncovered insights into AI-assisted commit patterns and developer workflows by implementing time series decomposition and cosine similarity analysis.
- Identified a significant structural shift post-2015, revealing that AI tools accelerated feature development while increasing message redundancy and decreasing documentation quality.